



9/1/2026

N + - Network Fundamentals & Building Networks.

TCP/IP Suite.

TASK 1.

❖ **Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.**

➤ **Wireshark Packet Identification:**

After refreshing the **Flipkart homepage** in Wireshark, I identified one packet for each layer by checking the **Protocol** and **header information**.

Layer	Packet / Protocol	Header Details	Purpose
Application	HTTP/HTTPS	HTTP request / TLS information	Handles communication between the browser and Flipkart website.
Transport	TCP	Source Port, Destination Port, Sequence Number	Provides reliable delivery of data between devices.
Internet	IP	Source IP, Destination IP, TTL	Provides addressing and routes packets across networks.
Network Access	Ethernet	Source MAC, Destination MAC, EtherType	Delivers frames between devices on the local network.

TASK 2.

❖ Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

➤ **Comparison of OSI and TCP/IP Models:**

OSI Model Layer	TCP/IP Model Layer	Real-World Example
7. Application	Application	WhatsApp message
6. Presentation	Application	Instagram image formatting
5. Session	Application	Zoom video session
4. Transport	Transport	WhatsApp message delivery using TCP
3. Network	Internet	Zomato order routed to its server
2. Data Link	Network Access	Wi-Fi frame sent from phone to router
1. Physical	Network Access	Wi-Fi radio signals carrying data

TASK 3.

❖ Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines, Look for fields like Source Port, Destination Port, Sequence Number, etc.

➤ TCP Segment Header Fields:

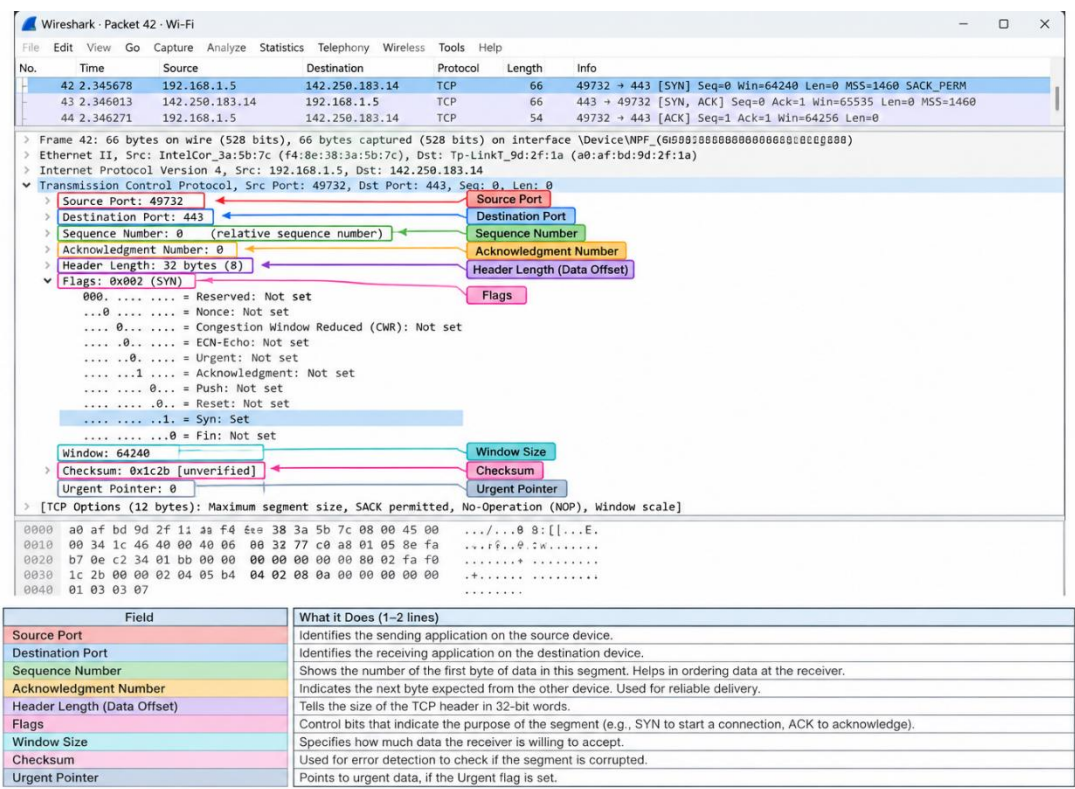


Figure : TCP Segment Header with Transport Layer Fields.

TCP Header Field	What It Does
Source Port	Identifies the application sending the data.
Destination Port	Identifies the application that should receive the data.

Sequence Number	Keeps track of the order of data so it can be correctly reassembled.
Acknowledgment Number	Indicates the next data expected from the other device.
Header Length	Shows the size of the TCP header.
Flags	Controls TCP communication, such as starting or acknowledging a connection.
Window Size	Shows how much data the receiver can accept at a time.
Checksum	Checks whether the TCP segment has been corrupted during transmission.
Urgent Pointer	Points to urgent data when the URG flag is used.

TASK 4.

- ❖ **Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).**

➤ **Message Flow Through TCP/IP Layers:**

Note: Diagram provided on the next page.

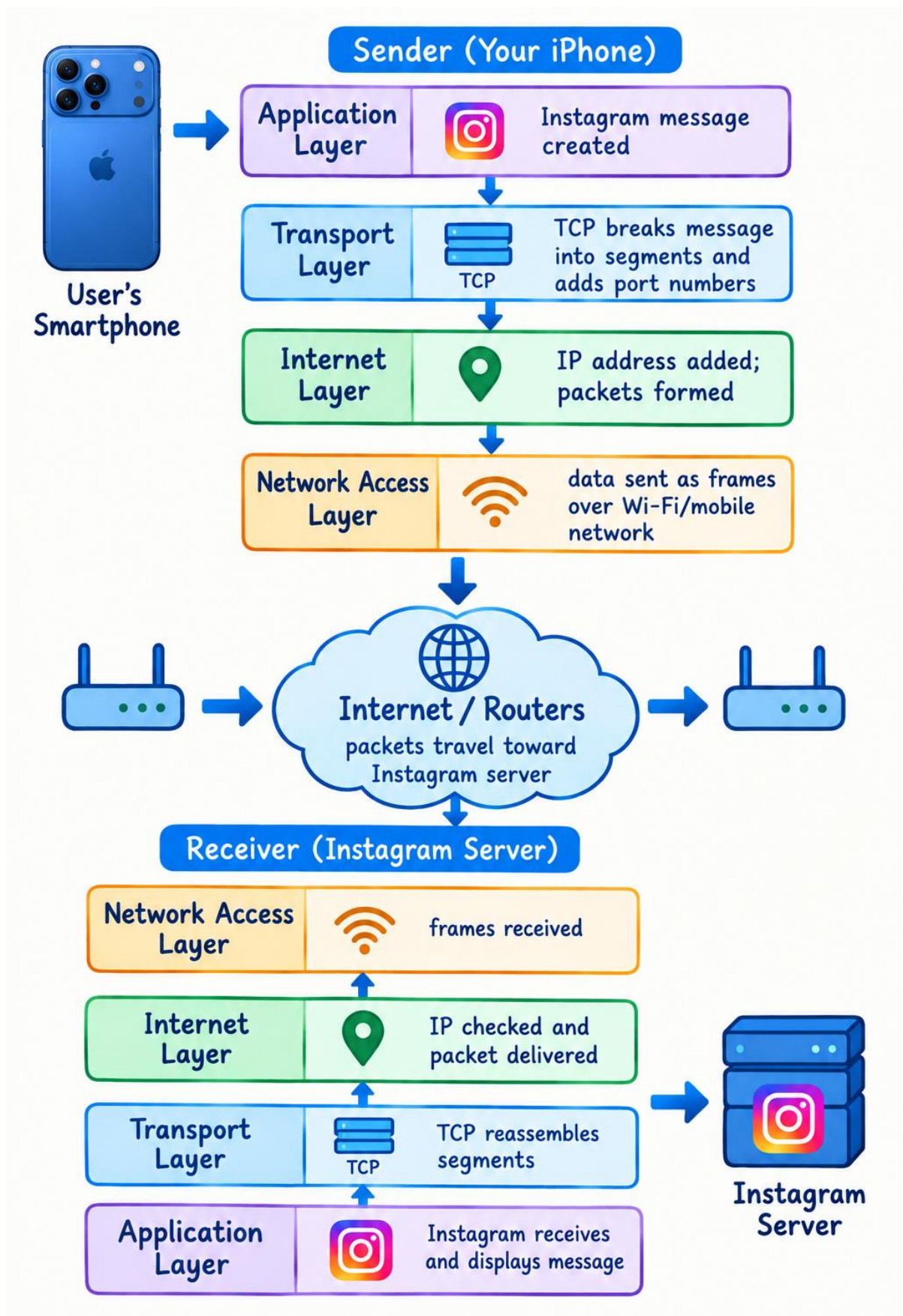


Figure : How an Instagram Message Travels Through the TCP/IP Layers.

🔗 Flow shown in the diagram:

- ➦ **Application Layer:** Instagram message is created.
- ➦ **Transport Layer:** TCP breaks the message into segments and adds port numbers.
- ➦ **Internet Layer:** IP addresses are added and packets are formed.
- ➦ **Network Access Layer:** Data is sent as frames over Wi-Fi/mobile network.
- ➦ **Instagram Server:** The data is received, reassembled, and the message is processed.

TASK 5.

❖ **Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.**

➤ Wireshark Packet Analysis:

Selected Packet: TCP

TCP/IP Layer: Transport Layer

🔗 How I Identified It:

I identified the packet as TCP by checking the Protocol column in Wireshark. The packet also showed Source Port and Destination Port, which are TCP header fields.

🔗 Real-World Action:

This packet represents the reliable transfer of data when sending a message on WhatsApp. TCP helps deliver the message data correctly between the device and the server.